Distance Metric for Similarity

- **Euclidean Distance**

$$d = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$

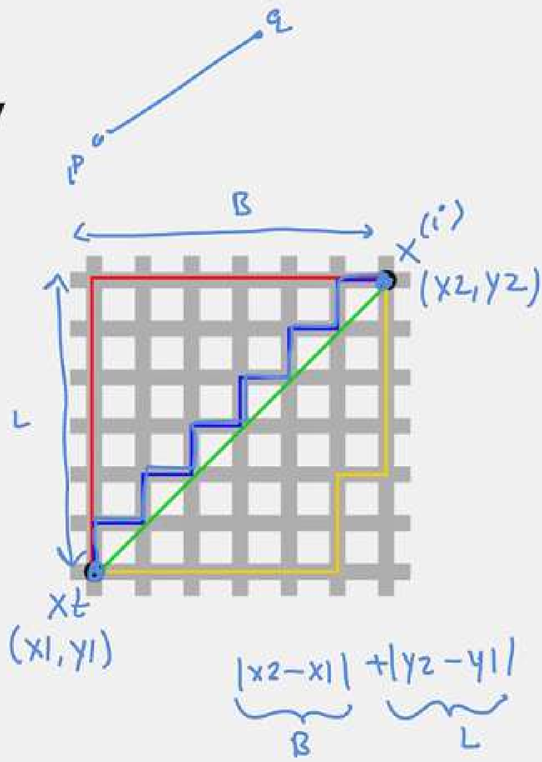
$q = (q_1, q_2, q_3 \cdots q_n)$

$p = (p_1, p_2 \cdots p_n)$

- **Manhattan Distance**

p, q

$$d_{mh} = |p_1 - q_1| + |p_2 - q_2| \cdots + |p_n - q_n|$$



k = 3, 5, 7, 11 ....

Choosing the right value of K.

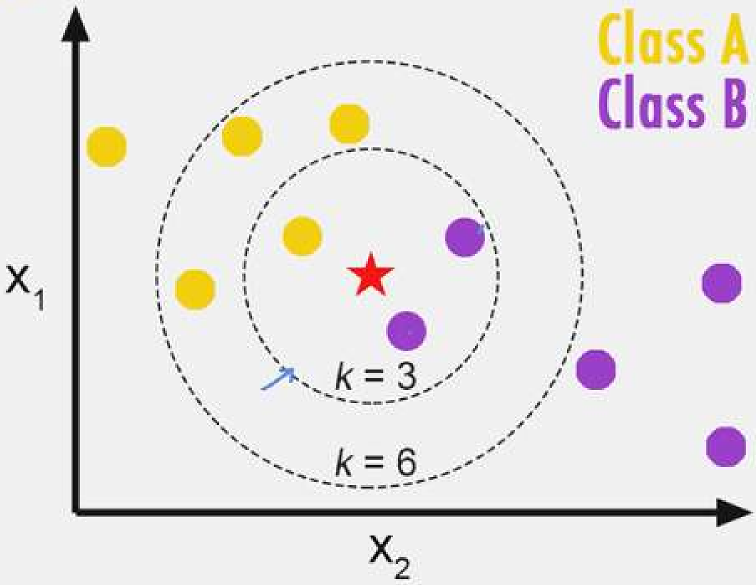
Even classes

2

K ⇒ odd

K = 3 label purble

k = 6 label yellow



k = 3

Age, Salary)

P → (20-50, 10,000 - 10 Lac)

$\boxed{\begin{matrix} \mu = 0 \\ \sigma = 1 \end{matrix}}$

$\uparrow d = \sqrt{(Age1 - Age2)^2 + (S1 - S2)^2}$

KNN requires scaling of data because KNN uses the Euclidean distance between two data points to find nearest neighbors. Euclidean distance is sensitive to magnitudes

Normalise

The features with high magnitudes will weight more than features with low magnitudes.

Advantages

1

k-NN doesn't require a pre-defined parametric function f(X) relating Y to X makes it well-suited for situations where the relationship is complex/non-linear to be expressed with a simple linear model.Hence, KNN can be useful in case of nonlinear data.

Training

There is no need to train a model for generalisation, that is why KNN is known as the simple and instance-based learning algorithm

Test

85%

80%

The model can be used for both classification and regression tasks and can act as good baseline model.

Neural Net

$\boxed{2GB}$

↑

Memory

$\{[\theta_0, \theta_1, \theta_2 \ldots \theta_n] \Rightarrow$ wts

0.01 MB

Disadvantages

Since the model isn't learning anything, you have to iterate over entire dataset for making every prediction. This makes testing phase KNN slow on larger datasets.

M = 10,000 training

= 500 test

(10,000 x 500)

Training Data

Costlier in terms of time and memory, requires storing entire dataset for memory during the prediction phase.